

TABLE VII: Lorentzian-fit parameters of ISGQR strength distributions in the Sn isotopes, as extracted from MDA. The results from TAMU work, where available, are provided for comparison [19, 21].

Target	E_{ISGQR} (MeV)	Γ (MeV)	EWSR ^a	Reference
¹¹² Sn	13.4 ± 0.1	7.0 ± 0.5	1.08 ± 0.04	This work
	$13.48^{+0.15}_{-0.14}$	$4.90^{+0.22}_{-0.27}$	$0.88^{+0.14}_{-0.13}$	TAMU
¹¹⁴ Sn	13.2 ± 0.1	6.8 ± 0.4	1.25 ± 0.05	This work
¹¹⁶ Sn	13.1 ± 0.1	6.4 ± 0.4	1.12 ± 0.04	This work
	13.50 ± 0.35	5.00 ± 0.30	1.08 ± 0.12	TAMU
¹¹⁸ Sn	13.1 ± 0.1	6.6 ± 0.3	1.08 ± 0.03	This work
¹²⁰ Sn	12.9 ± 0.1	7.0 ± 0.7	1.04 ± 0.04	This work
¹²² Sn	12.8 ± 0.1	7.8 ± 0.6	1.25 ± 0.04	This work
¹²⁴ Sn	12.6 ± 0.1	7.7 ± 0.9	1.13 ± 0.04	This work
	$12.72^{+0.11}_{-0.11}$	$4.20^{+0.32}_{-0.03}$	$0.89^{+0.15}_{-0.10}$	TAMU

^aOnly statistical uncertainties are included; systematic errors, mostly from DWBA calculations, are $\sim 20\%$.

that are significantly closer to the experimental values, in particular with the surface pairing forces and the SKM* interaction ($K_\infty \sim 215$ MeV) [78]. Thus, while pairing effects lower the ISGMR excitation energies, one still needs to reduce the K_∞ value by $\sim 10\%$ for achieving a reasonable agreement with the experimental data.

A very intriguing possibility in explaining the “softness” of the Sn isotopes has been offered very recently by Khan [79, 80]. The author asserts that, in analogy with the mutually-enhanced-magicity (MEM) effect observed in predictions of masses with different energy-density functionals [81, 82], the ISGMR energy in the doubly-magic nuclei might be anomalously higher. The obvious implication is that the calculations using interactions that are successful in describing the ISGMR in the doubly-magic nucleus ²⁰⁸Pb would necessarily overestimate the ISGMR energies in the open-shell nuclei. If this effect is manifested in any significant way, the energy of the ISGMR in the non-doubly-magic Pb isotopes, ²⁰⁴Pb and ²⁰⁶Pb, would be measurably lower than that in ²⁰⁸Pb [80]. In the only measurement of the ISGMR in ²⁰⁶Pb reported so far [83], this conjecture does not appear to hold. Still, precise